

Mathew H. Vallejo

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PROFESSIONAL PROFILE

Interdisciplinary PhD researcher at McGill University at the intersection of spatial audio, signal processing, machine learning, and psychoacoustics. Current dissertation work (SPIRAL: Spatial Impulse Response Archive for Learning) centered on developing generative models for spatial room impulse responses trained on geometric and multimodal inputs, with a cross-modal retrieval component linking acoustic outputs to perceptual language. Model outputs are evaluated via perceptual listening tests and objective acoustic metrics. Research applications include immersive audio, AR/VR, and content creation.

TECHNICAL SKILLS

SPATIAL AUDIO CAPTURE & REPRODUCTION

Multichannel/binaural/ambisonic recording techniques • Microphone array calibration and processing • Spatial room impulse response (SRIR) capture • Reverb auralization • Ambisonics encoding/decoding • Evaluation of spatial imaging, localization, externalization

DIGITAL SIGNAL PROCESSING (DSP)

Filter design and implementation • Time-frequency analysis • Beamforming and adaptive filtering • Feature extraction • Spatial domain analysis • Algorithm prototyping

MACHINE LEARNING FOR AUDIO

Neural network training (PyTorch, TensorFlow) • Multimodal learning (audiovisual) • Generative modeling • Dataset construction and preprocessing

PSYCHOACOUSTICS

Design and analysis of listening experiments • MUSHRA testing • Statistical analysis of perceptual data • Bias control and test validation following ITU-R BS.1534-3

ACOUSTICS & AURALIZATION

Room acoustics measurement/modeling • Acoustic simulation and convolution-based rendering • Sound field analysis • Perceptual evaluation of virtual acoustic environments

PROGRAMMING

Python (proficient) • MATLAB (proficient) • C++ (working knowledge)

PUBLISHED RESEARCH

[1] G. Grazioli, J. Kelly, M. Vallejo, R. King, and W. Woszczyk, "Investigation of spatial resolution of first and high order ambisonics microphones as capturing tool for auralization of real spaces in recording studios equipped with virtual acoustics systems," Audio Engineering Society Convention Express Paper, 157th Convention of the AES, New York, NY, Oct. 8–10, 2024. Available: <http://www.aes.org/e-lib>

EDUCATION

Ph.D., Sound Recording McGill University Montreal, QC, Canada <i>Dissertation:</i> SPIRAL: Spatial Impulse Response Archive for Learning	2023 – Present
M.M., Music Technology Berklee College of Music Valencia, Spain	2021 – 2022
Diploma, Music Engineering Berklee College of Music Boston, MA	2019 – 2021
B.A., Economics University of Puget Sound Tacoma, WA	2010 – 2014

PROFESSIONAL HISTORY

Student Coordinator (RA1)

May 2025 – Present

CIRMMT – Centre for Interdisciplinary Research in Music Media and Technology | Montreal, QC

- Support and assist with organization of interdisciplinary research and events for Research Axis 1: Instruments, devices, and systems
- Facilitate collaboration among student researchers, faculty, and distinguished lecturers at the intersection of acoustics, DSP, instrument design, immersive systems, and interactive music technologies

Course Instructor – Audio Fundamentals

July 2024 – Present

McGill University | Montreal, QC

- Lead lecturer for MUSR 200 undergraduate seminar (60+ students)
- Develop and maintain course materials for in-depth treatment of audio engineering theory

Research Assistant – Virtual Acoustics Technology

August 2024 – Present

McGill University – Immersive Media Lab (IMLab) | Montreal, QC

- Conduct research on Virtual Acoustics Technology (VAT)
- Design and implement software for impulse response auralization and real time reverb cancellation

Teaching Assistant – Electronics

August 2024 – Present

McGill University | Montreal, QC

- Support professors through preparation, instruction, and evaluation for undergrad and graduate-level coursework in analog electronics
- Guide students through theory and practice of circuit analysis/design and practical implementation

Sound Designer – Graduate Fellowship

July 2022 – May 2023

Earth Moments | London, UK

- Edited, compiled, and categorized audio samples for music production sample and loop packs
- Developed creative effects chains for raw audio processing in Ableton Live, Logic Pro X, and Kontakt

Mixing & Mastering Engineer

Sept 2021 – 2023

Pleasure Craft Studios | Montreal, QC

- Mixed and mastered across musical styles in stereo, multichannel, and Dolby Atmos formats

General Manager / Professional Musician

2015 – 2019

Pono Ranch, LLC & Self-Employed | Seattle, WA

- Managed operations, budget, booking, and staffing for one of Ballard's largest music venues (500–600 guests nightly)
- Concurrent work as professional guitarist and bassist